Evolution of Gene Network Analysis Methods: Towards an Approach Using Random Matrix Theory

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1- Introduction

Gene networks are among the most crucial frameworks to study in biological research today. These methods have evolved from their first application of network science in genomic biology and combined with ecological models in order to improve performance metrics like accuracy and interpretability over time.

Traditionally, the origins of network analysis in science go back as far as 1736 when Leonhard Euler solved the "Seven Bridges of Königsberg" problem that enabled development of graph theory and its applications[1]. The challenge posed to Euler was deceptively simple: Could a person cross all seven bridges exactly once without retracing their steps? Euler approached this problem by abstracting the geography of Königsberg into a network of nodes and edges. The landmasses were represented as nodes, and the bridges connecting them were represented as edges.

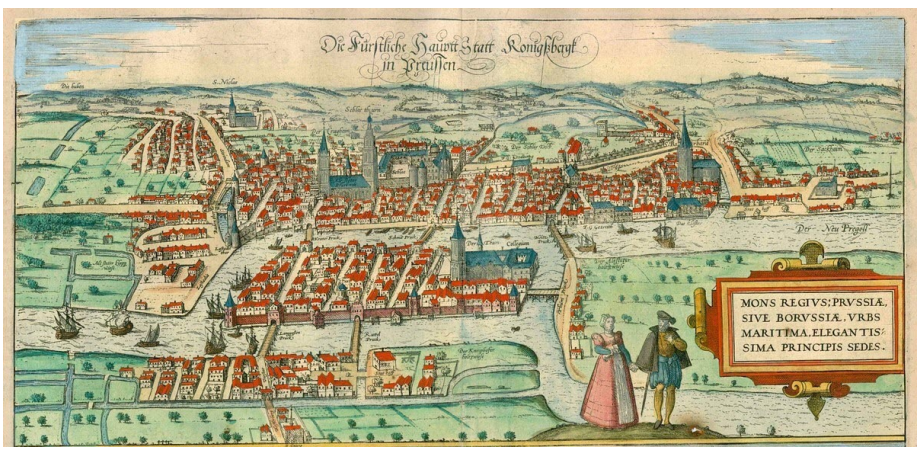

Figure 1: Seven Bridges of Königsberg[2]

Euler proved that the problem had no solution, laying down two key principles in the process:

1. Nodes and Edges: Euler identified that the ability to traverse a network depends on the degree of each node (the number of edges connected to it). For a path that crosses each edge exactly once (an Eulerian path), all but two nodes must have an even degree.

2. Graph Connectivity: The network must be connected, meaning all nodes must be reachable from any other node.

In the case of Königsberg, all four nodes had an odd degree, making it impossible to traverse the network under the stated conditions. This conclusion not only resolved the Königsberg problem but also established the first theorem of graph theory.

Jacob Moreno and Helen Jennings took this idea a step further in the 1930s, drawing social relationships with sociometric maps that would be some of the first systematic applications of network analysis to social science[3]. Use of network analysis has been applied to other domains like physics[4] and chemistry[5], showing how versatile and hwo impactful this field can be. This is why, in this work, the evolution on gene network analysis will be discussed, with a particular focus on the use of Random Matrix Theory (RMT) to increase robustness during network construction.

Gene networks are intricate representations of interactions among genes and their products within biological systems[6]. These networks, composed of nodes symbolizing genes and edges reflecting interactions[7], offer a system-wide perspective on cellular processes. Researchers leverage these networks to investigate critical biological phenomena[7], such as development[8], disease progression[9], and evolutionary adaptations[10]. Notably, gene networks are instrumental in identifying gene modules—highly connected clusters of genes that frequently correspond to functional units and hub genes, which play pivotal roles in maintaining cellular integrity[11].

The construction of gene networks employs diverse methodologies. Co-expression networks, for instance, utilize correlation matrices derived from gene expression data to identify links between genes that exhibit strong statistical relationships[11]. These approaches often rely on determining appropriate thresholds to distinguish meaningful biological connections from random noise, a process that remains subjective and heavily influenced by prior knowledge or statistical frameworks. Despite their utility, traditional network approaches face challenges such as scalability and sensitivity to noise, emphasizing the need for advanced methods to refine and automate the thresholding process[12].

This bibliography takes you on a journey through the history of methods for analyzing gene networks, with particular focus on the game-changing application of RMT in providing greater robustness to network construction. This work highlights the need for such advanced methods as well as the integration of approaches like co-expression networks in frameworks such as the Molecular Ecological Network Analysis Pipeline (MENAP) that aim to capture complex biological architectures.

2- Network approaches applied in genomic biology

A) Equation-based network methods

Equation-based methods offer a structured approach to modeling the dynamics of gene regulatory networks using ordinary differential equations (ODEs) to describe mRNA concentrations over time. Linearizing these ODEs around a steady-state point simplifies the analysis, enabling the representation of gene interactions through a connectivity matrix A, where a_{ij} quantifies the influence of gene j on gene i[12].
To capture the system's responses to external changes, perturbations (b_i) are incorporated into the ODE framework, providing a means to simulate environmental or experimental variations[13]. These perturbations allow researchers to explore the robustness and adaptability of the network.

Various methods have been employed to infer these networks:

- **Singular Value Decomposition (SVD):** SVD is utilized to decompose expression data into principal components, identifying sparse network patterns while reducing noise and computational complexity[13].
- **Robust Regression:** Combined with SVD, robust regression enhances the reconstruction of connectivity matrices by prioritizing sparsity and minimizing the impact of outliers[13].
- **Mutual Information and Boolean Networks:** Algorithms like REVEAL use mutual information to infer regulatory relationships based on input-output state transitions, suitable for binary models of gene activity[14].
- **Noise and Overdetermined Systems:** Designing experiments with $M > N$ perturbations (where M is the number of experiments and N is the number of genes) ensures robustness against noise. When this is infeasible, techniques like ridge regression or sparsity constraints become essential[13].

Despite their strengths, equation-based methods rely on assumptions such as the validity of the linear approximation, which may fail for large perturbations. Moreover, sparse network reconstruction demands careful experimental design to balance the number of perturbations with data quality[13].
These approaches provide powerful tools for inferring gene regulatory networks by integrating theoretical models with experimental data, enabling iterative refinements and deeper insights into biological systems.

B) Bayesian network methods

Bayesian networks are probabilistic graphical models that are used to analyze relationships between variables. These networks are particularly well-suited for genomic biology because they help in exploring complex interdependencies such as gene expression patterns and the dynamics of cancer progression[9,15].

A Bayesian network uses a directed acyclic graph (DAG) to model joint probability distributions, where nodes represent variables and edges reflect conditional relationships. Their design makes them perfect for modeling locally dependent systems. This allowing detailed exploration of processes such as gene regulation and mutation accumulation[9].

Learning a Bayesian network requires determining a structure that most effectively represents the observed data. This process often relies on statistical scoring functions, such as Bayesian or BDe scores, to evaluate potential network structures while balancing accuracy and simplicity. To address the computational challenges associated by high-dimensional datasets, such as those containing thousands of genes, methods like the Sparse Candidate algorithm restrict the search to a smaller subset of relevant candidate variables. This method enables rapid and resource-efficient algorithms[9].

These networks have proven to be valuable tools in various genomic biology applications:

- **Gene Expression Analysis:** Bayesian networks uncover gene interactions and transcriptional regulation mechanisms by analyzing statistical dependencies. By detecting Markov blankets (variables directly influencing a gene), they pinpoint variables with direct influence on genes and suggest causal associations. These networks are resilient to noisy data and can estimate confidence in findings through techniques such as bootstrapping.

- **Cancer Progression Modeling:** Conjunctive Bayesian Networks (CBNs) and Hidden CBNs (H-CBNs) model the accumulation of genetic mutations and their interdependencies, giving doctors clues about cancer progression. H-CBNs incorporate observation error models to account for technical noise, improving their robustness and biological relevance.

However, bayesian networks rely on handling of priors and assumptions. When working with small datasets, prior knowledge strongly influences the learning process. While these networks can infer causal relationships under the Causal Markov Assumption, such interpretations should be made cautiously and require other types of validation. For exemple, hybrid approaches that combine methods with clustering algorithms to learn models over "clustered" genes[9].
Bayesian networks powerful methods for addressing complex, high-dimensional problems in genomic biology, providing robust statistical analysis and efficient computational approaches to understanding gene regulation and disease progression.

C) Relevance/co-expression network method

The relevance/co-expression network method is an analytical framework designed to elucidate functional relationships among genes by investigating their co-expression patterns across diverse conditions or sample sets. This method begins with the calculation of pairwise correlations between gene expression profiles, commonly using Pearson correlation coefficients, which serve as a measure of similarity[11]. These correlations are then transformed into connection weights via an adjacency function, where soft-thresholding is preferred over hard-thresholding to retain biological nuances[16,6]. The resulting network comprises nodes, representing genes, and edges, reflecting the strength of their co-expression relationships. By applying a suitable threshold, weaker connections are excluded, enabling the identification of gene clusters, or modules, with significant co-expression[17].

Such modules often highlight genes involved in shared biological pathways, revealing insights into regulatory mechanisms and system-level gene interactions[18]. Relevance networks have been employed to link gene expression with phenotypic traits, such as drug susceptibility, offering hypotheses about gene roles in specific biological processes or responses to environmental stimuli[16]. This approach has demonstrated utility in contexts ranging from oncogenic signaling to evolutionary studies of co-expression networks across species[6]. The methodology thus integrates genomic data into functional networks, bridging the gap between molecular data and systems biology.

Among the various methods for constructing gene co-expression networks, the correlation-based relevance network method stands out for its simplicity and resilience to noise[16]. This method calculates pairwise correlations among genes and uses thresholds to filter out weak associations, producing networks that are straightforward to interpret[17]. However, despite its advantages, the reliance on arbitrary thresholds is a significant limitation. These thresholds, often chosen based on subjective judgment or convenience, can introduce bias and affect the reproducibility and objectivity of the resulting networks[6]. Arbitrary thresholding not only impacts the detection of biologically relevant interactions but also raises concerns about the method's capacity to reflect the true complexity of gene regulatory mechanisms[19]. Addressing these limitations requires more systematic approaches to threshold selection. Such advancements would enhance the robustness and reliability of relevance network analyzes, bridging the gap between simplistic correlation-based methods and more sophisticated, biologically accurate models.

D) General comparaison

When comparing these methods, several key aspects become evident. Bayesian methods are particularly notable for their robustness due to their probabilistic nature, while equation-based approaches are more sensitive to noise without careful experimental design. For scalability, Bayesian and relevance methods are better suited for handling large datasets, whereas equation-based methods may encounter difficulties. Regarding biological accuracy, relevance methods can fall short because of their reliance on simplistic correlation metrics, potentially overlooking nuanced interactions. Finally, in terms of ease of interpretation, relevance networks are the most straightforward, followed by Bayesian networks. Equation-based models, although powerful, are the most complex and challenging to interpret due to their mathematical intricacies.

Because of its computational simplicity and the nature of microarray data (typically noisy, highly dimensional and significantly under-sampled)[19], correlation-based relevance network method are most commonly used for identifying cellular networks. It is important to address the limitations of arbitrary thresholding so those network methods could provide a more comprehensive and biologically accurate representation of gene interactions. This is exactly what MENA does, by integrating RMT to provide a more systematic and robust approach to threshold selection.

3- Random Matrix Theory

A) Fundamentals of Random Matrix

Random Matrix Theory bridges linear algebra and probability theory[20], examining the statistical behavior of matrices with randomly distributed elements. Originally introduced by Wigner and Dyson in the 1960s to study the spectral properties of complex nuclei, random modeling has since been applied to identifying and analyzing phase transitions linked to disorder and noise[12]. It allows to find order in chaos, revealing underlying structures in complex systems like co-expression networks[21], financial markets[22], and quantum physics[23]. A key focus of RMT is to understand the properties of their eigenvalues of the matrices with random entries.

TA basic calculation in RMT is finding the spacing distribution of eigenvalues. A simple example involves a 2x2 real symmetric matrix with Gaussian random variables as entries.

Consider the matrix X:

$$X = \begin{pmatrix} x_1 & x_3 \\ x_3 & x_2 \end{pmatrix}$$

$$x_1, x_2 \sim N(0,1) \quad x_3 \sim N(0, \frac{1}{2})$$

Where $N(0,1)$ denotes a Gaussian distribution with mean 0 and variance 1 and $N(0, \frac{1}{2})$ denotes a Gaussian distribution with mean 0 and variance $\frac{1}{2}$.

The variance of the off-diagonal element is chosen to be half of the variance of the diagonal elements for a specific reason that allows an ea The question is that we want to find the probability density function (pdf) of the spacing s between the two eigenvalues. So $s = \lambda_1 - \lambda_2$ where λ_1 and λ_2 are the eigenvalues of the matrix X.
This spacing for a 2x2 matrix can be calculated like this:

$$s = \lambda_1 - \lambda_2 = \sqrt{(x_1 - x_2)^2 + 4x_3^2}$$

We're gonna skip the whole demonstration of the calculation, but the final equation for the pdf of the spacing s is:

$$P(s) = \frac{s}{2} e^{-\frac{s^2}{4}}$$

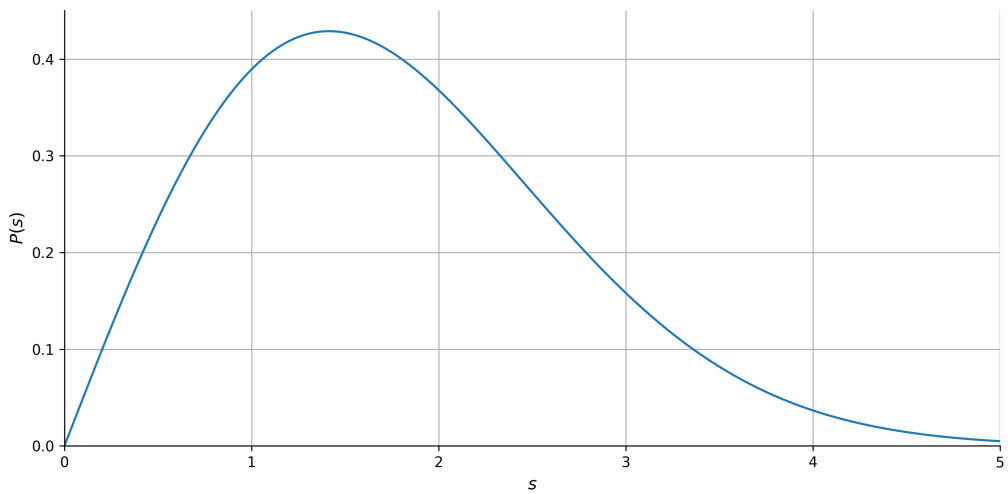

Figure 2: Wigner's surmise

Despite its simplicity, this result is remarkably profound: it reveals that the probability of sampling two eigenvalues that are "very close" to each other (as $s \to 0$) is extremely low. It's as though each eigenvalue "senses" the presence of the others and adjusts to maintain a certain distance—neither too close nor too far. This behavior is reminiscent of birds perched on an electric wire or cars parked along a street: maintaining a balance between proximity and spacing.[24]

Random Matrix Theory predicts two universal extreme distributions for the nearest neighbor spacing distribution (NNSD) of eigenvalues: the Gaussian Orthogonal Ensemble (GOE) statistics, reflecting the random characteristics of complex systems, and the Poisson distribution, representing system-specific, nonrandom properties of complex systems. This transition highlights level repulsion, where eigenvalues tend to "avoid" close proximity. The phenomenon underscores intrinsic correlations among eigenvalues, even when matrix entries are independently distributed.
By investigating eigenvalue spacing distributions, researchers identify key properties like the Wigner's surmise distribution in GOE systems or the exponential decay of Poisson distributions in decoupled systems. These techniques offer a powerful framework for exploring complex systems, including biological networks and beyond.[25]

Random Matrix Theory allow us to have a better understanding of the organization of complex systems by analyzing eigenvalue spacings, capturing transitions from global interactions to modular structures. This approach underscores the utility of statistical models like Wigner's surmise and Poisson distributions in exploring biological networks and other interconnected systems.

B) Applications of Random Matrix Theory in Many-body Systems

By describing the statistical properties of spectra in complex quantum systems, RMT bridges seemingly disparate phenomena through its universal principles. The role of RMT in understanding many-body systems, its implications for quantum chaos, and its connections to field theory and statistical mechanics, allow us to understand the versatility and foundational importance of it in modern physics.

Many-body systems encompass complex structures involving numerous particles interacting via two-body forces. Examples include atomic nuclei, which consist of nucleons bound by strong nuclear forces, and atoms and molecules, where ions and electrons interact through electromagnetic forces. These systems demonstrate high levels of complexity. The Hamiltonian is an operator which determines the evolution of a quantum state through the Schrödinger equation. It's described for N particules like :

$$\hat{H} = \sum_{n=1}^{N} \hat{T}_n + \hat{V}$$

Where $\hat{T}_n$ is the kinetic energy operator of particle n and $\hat{V}$ is the potential energy function. The

key idea is to replace the complex, specific Hamiltonian of the system with an ensemble of random matrices that share the same symmetries.

At low incident energies, the use of the GOE in modeling compound nucleus scattering assumes that the nucleus equilibrates internally faster than it decays. However, as incident energy increases, the decay time becomes comparable to the equilibration time, meaning the nucleus can decay before full equilibration.
To address this, the model is extended using the nuclear shell model, dividing the compound nucleus into classes of states with fixed particle-hole numbers. Each class is represented by a random matrix, and the coupling between neighborefers to the absence of a fermion in an energy level it would occupy in the ground statering classes, governed by the two-body interaction, is also modeled using random matrices. Imagine sorting all possible quantum states of a system (like a nucleus) into different groups based on their particle-hole number. This means each group contains states with the same number of particles excited above the ground state and the same number of holes left behind. Each of these groups is then modeled using a random matrix.

Instead of trying to calculate the exact energy values for each state in a group, which is extremely slow and complex and laborious, we use a random matrix to represent the overall statistical behavior of the energy levels within that group. The random matrix is chosen from an ensemble that respects the symmetries of the physical system. The total Hamiltonian is then a band matrix whose entries are random matrices. By doing this, RMT provides a deeper understanding of the behavior of resonances and the fluctuations in cross-sections within models of many-body systems.

4- Molecular Ecological Network Analysis

A) Pipeline construction

Molecular ecological networks (MENs) represent biological interactions within microbial communities, where nodes symbolize molecular markers such as operational taxonomic units (OTUs), functional genes, or intergenic regions, and edges denote the interactions between them. These networks are categorized into functional molecular ecological networks (fMENs), derived from functional gene markers, and phylogenetic molecular ecological networks (pMENs), based on phylogenetic gene markers.

The process of Molecular Ecological Network Analysis (MENA) is divided into two primary phases.

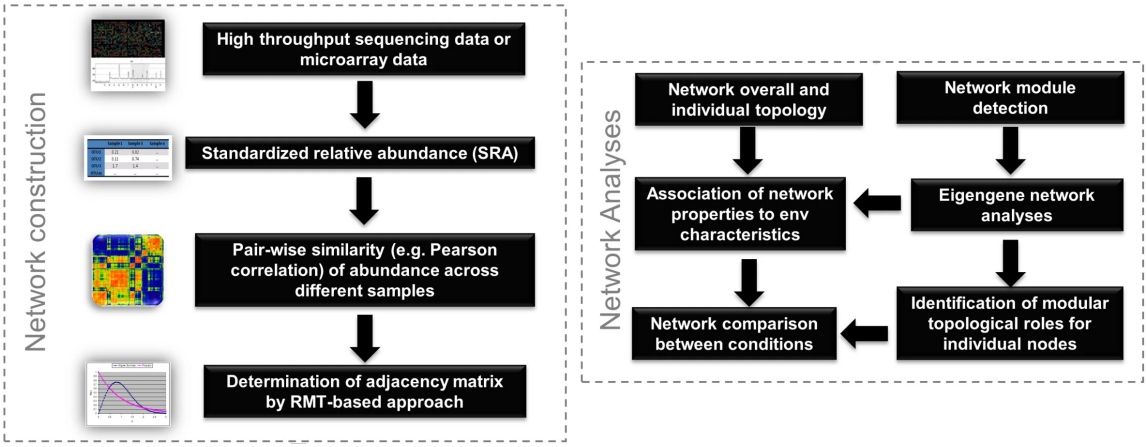

Figure 3: Overview of the Random Matrix Theory (RMT)-based molecular ecological network analysis[12]

The first phase is network construction, which involves data collection of data then its transformation or standardization to normalize, calculation of pairwise similarity matrices, and the determination of the adjacency matrix using Random Matrix Theory. The RMT-based approach is crucial for constructing an accurate network by defining connections between nodes based on statistical thresholds, resulting in an undirected network graph.

The second phase is network analysis, which includes network topology characterization to assess the overall structure and properties of the network, module detection to identify groups of tightly connected nodes known as modules. Then a module-based eigengene analysis to understand underlying patterns and functions, and the identification of modular roles to determine the importance and function of nodes within modules. Additionally, eigengene network analysis explores higher-order organizational structures within the network, and associations between network properties and environmental characteristics are established to understand environmental influences. Finally, comparative analysis evaluates network differences under varying conditions to assess how environmental changes affect network structure and interactions.

Collectively, these methods enable researchers to uncover the complex interactions within microbial ecosystems, identify key functional populations at the OTU level, and understand how environmental factors influence these networks.

B) Determination of the Adjacency Matrix Using Random Matrix Theory

RMT is utilized in MENA as a way to automatically identify thresholds for network construction. It is able to do that by examining the statistical properties of matrices derived

from high-throughput ecological data.

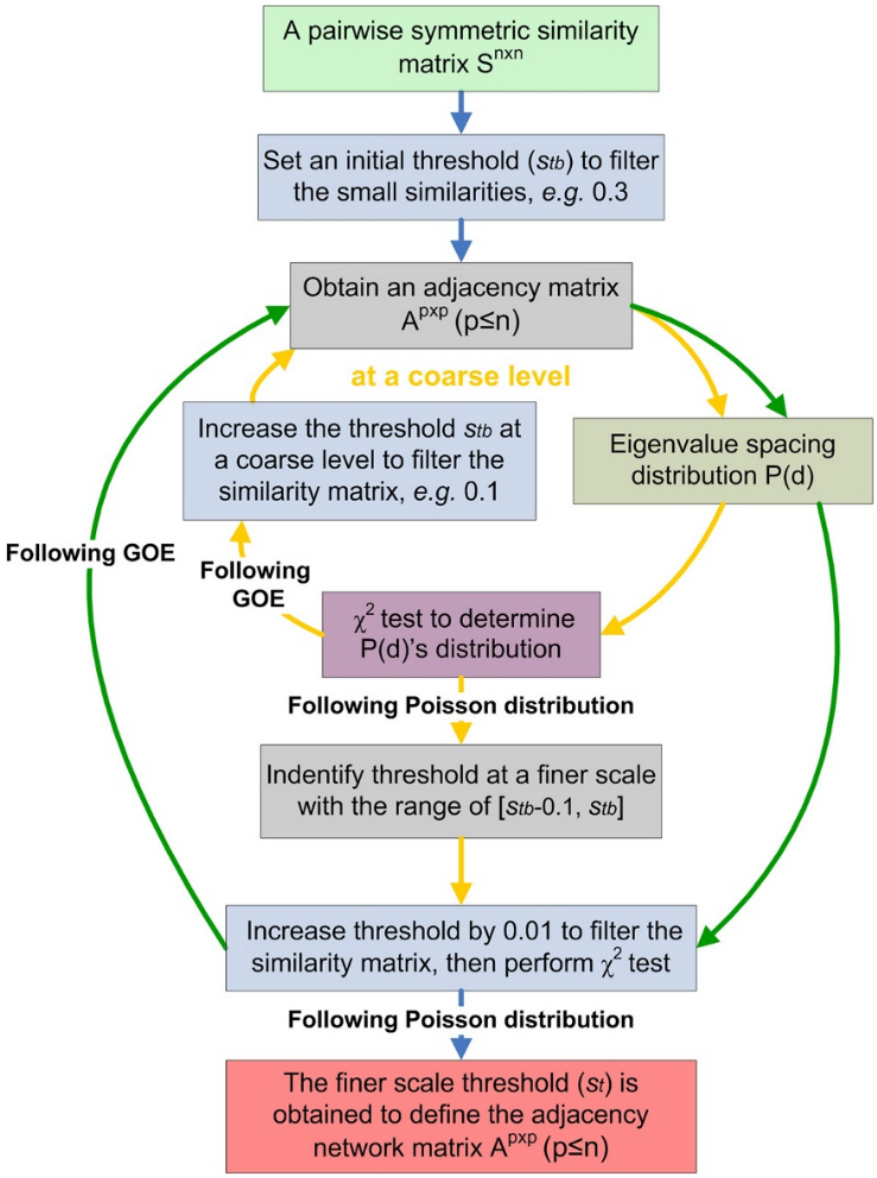

Figure 4: Process of random matrix theory-based approach for automatically detecting a threshold to construct molecular ecological networks[12]

At first, we have to compute the Pearson correlation matrix R^{nxn} using the standardized relative abundances of OTUs X^{nxm}, where n is the number of distinct OTUs and m is the number of samples. This matrix R^{nxn} is rapidly transformed into a similarity matrix S^{nxn} by just taking

the absolute values of R^{nxn}. An adjacency matrix A^{pxp}, where p is the number of OTUs retained in the adjacency matrix with non-zero rows or columns, is then defined according to a threshold s_{tb} set at first at 0.3.

For the completely uncorrelated eigenvalues, P(d) follows Poisson statistic, and it can be expressed by, $P(s) = e^{-d}$ and the correlated eigenvalues, P(d) closely follow Wigner-Dyson distribution of the GOE statistics, and it can be expressed by $P(s) \approx \frac{\pi s}{2} e^{-\frac{\pi}{4}s^2}$.

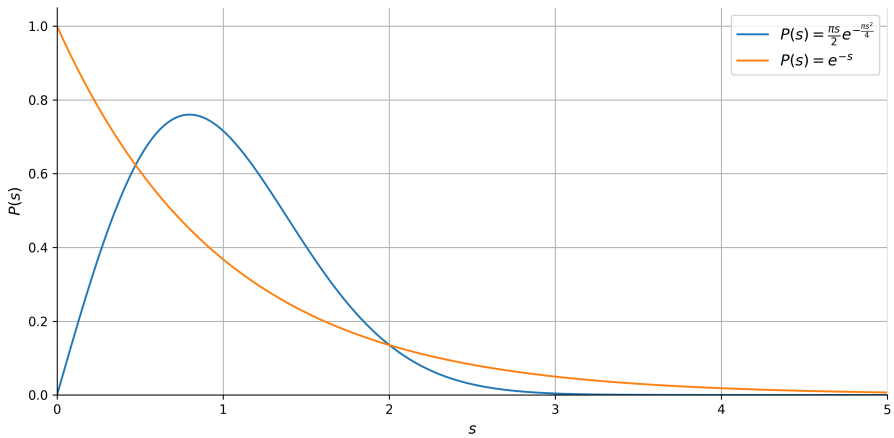

Figure 5: Wigner-Dyson and Poisson distribution

To determine whether the Nearest Neighbor Spacing Distribution follows the Wigner-Dyson distribution or the Poisson distribution, we use the chi-squared test to fit it to the Poisson distribution. By establishing the null hypothesis H_0 that $P(d)$ follows a Poisson distribution, we test if the NNSD conforms to this distribution. If the NNSD does follow a Poisson distribution, we then subtract 0.1 to the current threshold and then increase the threshold incrementally by 0.01 instead of 0.1.
After determining the final threshold value s_t at a finer scale, an adjacency matrix is constructed by retaining all OTUs with abundance similarity values exceeding the defined threshold. In the later network topological analysis, we have adopted an unweighted network. Therefore, the final adjacency matrix will be defined like $A^{pxp} = [a_{ij}]$ is:

$$a_{ij} = \begin{cases} 1 & \text{if } s_{ij} \geq s_t, \\ 0 & \text{if } s_{ij} < s_t. \end{cases}$$

C) Comparison to Legacy Network Methods

5- Conclusion

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Abstract

Gene networks serve as essential frameworks for many problems in modern biology, and it is increasingly clear that we need to understand their interactions in a systematic manner. The use of network science in genomic biology was the first application and has since evolved to include the basic understanding gained from ecology on whatever model has been envisaged, with increasing challenges for both accuracy and interpretability. Network analysis roots in science started in the 18th century with Leonhard Euler solving the "Seven Bridges of Königsberg" problem in 1736, creating a foundational concept for graph theory. In this bibliography, we will walk through the history of gene network analysis methods, pinpointing the turning point of using Random Matrix Theory (RMT) to increase robustness during network construction. Through examining (co-)expression networks to considering their incorporation into a framework like the Molecular Ecological Network Analysis Pipeline (MENAP), this work emphasizes the need for more powerful methods for capturing the complex structure of biological systems.

Keywords

Gene Network, Random Matrix Theory, Molecular Ecology, Network Analysis